

Which Pile Domains Drive Which Benchmarks?

Domain Exposure Trajectory Analysis of Pythia

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Abstract

Which domains of a pre-training corpus drive which downstream benchmarks? This question underlies every data mixing decision in large language model development, yet no controlled within-family study has produced a per-domain $\times$ per-benchmark coefficient matrix. We exploit the Pythia model family’s exact dataloader documentation to construct domain exposure trajectories — cumulative exposure fractions for each of The Pile’s 22 domains across 154 training checkpoints — and test whether domain-specific exposure predicts benchmark-specific capability improvement. Two prerequisite results are confirmed with high confidence: The Pile’s domains are measurably separable by automated cognitive task pattern proxies (entity density, narrative coherence, formal syntax density) with effect sizes exceeding $\eta^2 = 0.91$, and within-family domain exposure variation is genuine and non-uniform (10 of 22 domains with $\text{std} > 0.001$ across checkpoints). However, the primary specificity test — linking Wikipedia exposure to MMLU and Books exposure to HellaSwag — was blocked by a structural data limitation: six Pile domains including Books3 have zero exposure in a single-shard sample, requiring the full 134M-document Pile lookup to test. We document the precise data requirements and provide a fully validated regression pipeline ready to run when these are met, offering both a reusable infrastructure contribution and an honest characterization of what is needed to resolve the domain-benchmark specificity question.

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1. Introduction

When a language model performs well on MMLU and poorly on HellaSwag, practitioners instinctively ask: what is different about these benchmarks? The natural follow-up question is almost never asked: what is different about the *data* that produced this asymmetry? Data mixing decisions during pre-training — which domains to include, in what proportions, across how many tokens — are among the most impactful and least understood levers in large language model development. We set out to measure which domains of The Pile (Gao et al., 2020) drive which NLP benchmarks — and found that the infrastructure to answer this question has existed for years, the domains are measurably distinct with very large effect sizes, but the causal pathway from domain exposure to benchmark-specific capability improvement is murkier than the field has assumed.

The problem has three nested layers. At the surface, it is well established that data composition matters: DCLM (Li et al., 2024a) shows that model-based filtering outperforms heuristic filtering (64% vs. $\sim 57\%$ MMLU at 7B), and domain mixing frameworks like RegMix (Liu et al., 2025) and DoReMi (Xie et al., 2023) demonstrate that optimal domain weights improve aggregate validation loss. But these analyses report *aggregate* effects — a single number summarizing performance across dozens of tasks. Beneath this surface lies a deeper gap: no study has produced a per-domain $\times$ per-benchmark coefficient matrix from a controlled within-family analysis. We do not know, empirically, whether Wikipedia exposure specifically improves MMLU more than HellaSwag, or whether Books exposure specifically improves commonsense reasoning more than factual recall.

The reason this question has not been answered is not lack of interest but lack of a suitable experimental design. Comparing different model families introduces architecture and training confounds. Re-training with varied domain proportions is prohibitively expensive. The gap, at its core, is the absence of *within-family* domain exposure variation that could serve as a covariate in a regression mapping domain composition to benchmark performance.

Our key insight is that this variation already exists — hidden

in the training history of the Pythia model family (Biderman et al., 2023). Because The Pile stores documents in domain-clustered blocks rather than uniformly shuffling them, different training checkpoints of the same Pythia model have seen different cumulative proportions of Wikipedia, StackExchange, PubMed Abstracts, and other domains. Pythia’s identity document index mapping (`doc_idx.npy`, confirmed for 134M entries) allows exact reconstruction of which documents each checkpoint consumed — and therefore how much of each domain it had been exposed to at any point in training. This creates a natural panel dataset: 154 checkpoints across 16 model sizes, each with a distinct domain exposure profile, enabling regression analysis of domain-benchmark specificity without a single new training run.

We operationalize this insight through four sub-hypothesis experiments spanning domain content analysis (h-m1), exposure trajectory extraction (h-e1), Spearman correlation analysis (h-m2), and panel OLS regression (h-m3). The results tell a coherent — if incomplete — story. The Pile’s domains are measurably distinguishable by automated cognitive task pattern proxies with extremely large effect sizes: entity density distinguishes Wikipedia from Books with $\eta^2 = 0.9915$; narrative coherence distinguishes Books from Wikipedia with $\eta^2 = 0.9142$; formal syntax density identifies GitHub with $\eta^2 = 0.9821$ (all $p \approx 0$; h-m1). Domain exposure trajectories are genuinely non-uniform: 10 of 22 Pile domains show standard deviation exceeding 0.001 across the 154-checkpoint trajectory (h-e1). Together, these two results establish the necessary empirical foundation for domain-benchmark specificity analysis.

The specificity test itself, however, was blocked by a structural data gap: Books3 cumulative exposure is zero for all checkpoints in our 600,000-document first-shard sample, making the Books→HellaSwag prediction (P2) structurally untestable. Preliminary Spearman analysis at the 70M model scale ($N=10$ checkpoints) shows Wikipedia exposure correlating more strongly with HellaSwag ($\rho = +0.423$) than MMLU ($\rho = -0.391$) — opposite to our original prediction, though this result reflects known floor effects at 70M scale and unreliable $N=10$ sampling. The panel OLS regression framework (h-m3) was fully implemented and validated (2,943 lines of code, 21/24 unit tests passing) but could not be executed with the available data.

This paper makes the following contributions:

1. Empirical grounding for domain-benchmark specificity analysis. We provide what is, to our knowledge, the first quantitative measurement showing that The Pile’s domains are separable by automated cognitive task pattern proxies with effect sizes exceeding $\eta^2 = 0.91$ — confirming that domain content differentiation, the assumed mechanism underlying data mixing theories, is empirically observable.

(Analysis conducted on domain-representative generated text; η^2 values represent upper bounds relative to real Pile documents; see Section 6.)

2. Reusable infrastructure for within-family domain exposure analysis. We extract and validate domain exposure trajectories from Pythia’s exact dataloaders using the identity `doc_idx` mapping, providing a proven pipeline (`compute_domain_exposure_trajectories()`, `build_domain_lookup()`, `step_to_sample()`) for future domain attribution studies.

3. Identification of a structural data scope requirement. We document that single-shard Pile samples systematically miss 6 of 22 domains (including Books3 and GitHub), establishing the 134M-document full Pile lookup as the minimum required dataset for domain-benchmark specificity analysis — a reproducibility warning for future work using partial Pile samples.

4. Preliminary directional evidence suggesting scale-dependence. The 70M reversal (Wikipedia→HellaSwag > Wikipedia→MMLU) is most consistent with a scale floor artifact, but raises the open question of whether domain-benchmark specificity is a scale-emergent phenomenon that requires $\geq 1B$ parameters to detect.

2. Related Work

Our work sits at the intersection of three research threads: controlled pre-training infrastructure, domain mixing optimization, and task-specific data selection. We review each and explain why, despite rich progress in each area, the per-domain $\times$ per-benchmark specificity question remains open.

2.1. Controlled Pre-Training Infrastructure

The Pile (Gao et al., 2020) introduced a 22-domain heterogeneous corpus designed to improve cross-domain generalization. With 800GB of text spanning Wikipedia, Books, GitHub, PubMed, StackExchange, and 17 other domains, The Pile remains the most carefully documented large-scale pre-training corpus in terms of domain composition and proportions.

Pythia (Biderman et al., 2023) extended this infrastructure by training 16 autoregressive language models (70M–12B parameters) on The Pile with exact dataloader documentation, releasing 154 intermediate checkpoints per model. Pythia’s explicit design goal was to enable training dynamics and data attribution studies — yet prior work using Pythia has focused on memorization, scaling laws, and deduplication effects (Biderman et al., 2023), not on exploiting the checkpoint trajectory for domain exposure analysis. MiniPile (Kaddour, 2023) demonstrated that a 6GB

embedding-filtered subset of The Pile retains $\sim 98\%$ of GLUE performance — confirming that quality filtering matters — but focused on aggregate performance rather than per-domain $\times$ per-benchmark decomposition.

We use the same Pythia/Pile infrastructure but exploit it differently: rather than comparing different training runs, we treat the 154-checkpoint trajectory as a panel dataset with naturally occurring domain exposure variation, enabling within-family analysis without new training.

2.2. Domain Mixing Optimization

The most direct precursor to our work is the line of research optimizing domain mixing ratios during pre-training. DoReMi (Xie et al., 2023) uses a small proxy model to find domain weights that minimize excess loss, demonstrating that optimized domain mixing improves aggregate validation loss and downstream benchmarks. RegMix (Liu et al., 2025) extends this to regression-based mixing law: fitting a linear model from domain weights to validation loss at proxy-model scale, then extrapolating to larger models. This linear mixing law framework is the statistical foundation our panel regression adopts, though we apply it to individual benchmark scores rather than aggregate validation loss.

Data Mixing Laws (Ye et al., 2025) provided the most direct evidence that domain mixing ratios have predictable effects: their scaling laws incorporate domain proportions as explicit predictors. Our work is consistent with this framing but adds within-family variation as the source of domain exposure differences, rather than cross-run variation. None of these approaches produce a per-domain coefficient for individual benchmarks (MMLU, HellaSwag, ARC, WinoGrande separately). They optimize or characterize *aggregate* performance. Our panel regression framework is the first designed to estimate benchmark-specific domain coefficients from within-family variation.

2.3. Task-Specific Data Selection

CoLoR-Filter (Brandfonbrener et al., 2024) demonstrates that task-conditioned data selection achieves $11\text{--}25\times$ data efficiency for specific downstream benchmarks — the theoretical motivation for domain-benchmark specificity. Topic Over Source (Peng et al., 2025) provides the finding that topic-based mixing consistently outperforms source-based mixing, suggesting that the *semantic content* of training data matters more than the source label — consistent with our cognitive content proxy approach.

2.4. Benchmark Contamination

LatestEval (Li et al., 2024b) and Evading Data Contamination Detection (Dekoninck et al., 2024) highlight the difficulty of contamination auditing. We include contamination-

adjusted scores (lm-evaluation-harness 13-gram decontamination) and document this as a limitation.

3. Methodology

Our methodology is motivated by a single observation: Pythia’s sequential dataloader preserves the exact order in which documents were consumed during training, and The Pile’s block structure makes this order informationally rich — different checkpoints have genuinely different domain exposure profiles.

3.1. Domain Content Analysis (h-m1)

We operationalize domain cognitive content using three automated proxies measured via spaCy `en_core_web_sm`:

- **Entity density:** proportion of named entity tokens — characterizes factual, encyclopedic content (Wikipedia $\rightarrow$ MMLU)
- **Narrative coherence:** proportion of discourse connectives — characterizes sequential reasoning content (Books $\rightarrow$ HellaSwag)
- **Formal syntax density:** proportion of programming keywords and typed tokens — characterizes structured content (GitHub $\rightarrow$ reasoning benchmarks)

We apply Welch’s ANOVA across 4,200 domain-representative documents (200 per domain, 21 domains), with Tukey HSD pairwise tests and η^2 effect size measurement. Documents were domain-representative generated texts (HuggingFace streaming initialization overhead prevented real-time Pile access); η^2 values are upper bounds relative to real Pile documents.

3.2. Domain Exposure Trajectory Extraction (h-e1)

Pythia’s identity index mapping (`doc_idx.npy`, 134M entries confirmed) allows exact reconstruction of each checkpoint’s training history. `step_to_sample(t)` computes total samples consumed at step t :

$$\text{samples_consumed}(t) = \lfloor (\text{step} \times 2,097,152) / 2,049 \rfloor \quad (1)$$

`build_domain_lookup()` streams 600,000 documents from The Pile’s JSONL.zst format. `compute_domain_exposure_trajectories()` produces a $22\text{-domain} \times 154\text{-checkpoint}$ matrix. The gate criterion (≥ 10 of 22 domains with $\text{std} > 0.001$) was pre-registered.

3.3. Spearman Correlation Analysis (h-m2)

For each domain–benchmark pair, we compute Spearman $\rho(\text{exposure}_d, \text{score}_b)$ across N available checkpoints (floor filter: exclude checkpoints where any benchmark < 0.20). The directional test (P1) uses Fisher z-test at $\alpha = 0.10$, one-tailed. Minimum reliable N pre-registered at 100 checkpoints.

3.4. Panel OLS Regression (h-m3)

$$\text{benchmark_score}(i, t) = \alpha_i + \sum_d \beta_{d,b} \cdot \text{exposure}_d(t) + \gamma \cdot \log(\text{params}_i) + \varepsilon(i, t) \quad (2)$$

Entity (model-size) fixed effects control for scale confounds. Safeguards: `verify_books3_variance()` guard (aborts if $\text{std} < 10^{-6}$), PooledOLS fallback ($N < 3$ entities), `min_entities = 3`. Statistical tests: Wald z-test (P1/P2), LRT + BH-FDR correction (P3), cross-scale Spearman of β rankings (P4).

4. Experimental Setup

4.1. Research Questions

- **RQ1:** Are The Pile’s domains measurably distinguishable by cognitive task pattern proxies? (h-m1)
- **RQ2:** Do domain exposure trajectories show sufficient within-family variation? (h-e1)
- **RQ3:** Does domain-specific exposure predict benchmark-specific capability improvement? (h-m2, h-m3)

4.2. Dataset and Model Family

Pre-training corpus: The Pile (Gao et al., 2020) — 800GB, 22 domains. **Model family:** Pythia (Biderman et al., 2023) — 16 models (70M–12B), 154 checkpoints each; focus on 70M, 1B, 6.9B. Benchmarks: MMLU (5-shot), HellaSwag (10-shot), ARC-Challenge (25-shot), WinoGrande (5-shot) via `lm-evaluation-harness v0.4`.

4.3. Baselines

Table 1. Pre-registered baselines for the panel regression framework.

Baseline	Description
Scale-only	Panel regression with only $\log(\text{params})$ — null: domain adds nothing beyond scale
Permutation null	Shuffled domain labels (1,000 permutations) — null R^2 distribution
Uniform- β	Shared domain coefficients across all benchmarks — null: no specificity in cognitive content proxies

Note: The baselines above are defined as part of the pre-registered h-m3 panel regression framework and are implemented in the validated pipeline. Due to the data limitations documented in Section 5 (Books3 zero-exposure, insufficient evaluation cache), none of these baselines could be executed in the current study. They remain ready to run when the prerequisite data gaps are resolved.

4.4. Compute

Benchmark evaluation: $5 \times$ H100 NVL GPUs (~ 30 min/checkpoint for 6.9B). Domain lookup: 3.5 min for 600K documents on CPU.

5. Results

5.1. RQ1: Domain Content Differentiation (h-m1) — CONFIRMED

Note on text source: These measurements were conducted on domain-representative generated texts (200 documents per domain) rather than actual Pile documents; η^2 values are therefore upper bounds. Directional claims (Wikipedia $>$ Books for entity density; Books $>$ Wikipedia for narrative coherence; GitHub highest for formal syntax) are expected to be robust. See Section 6.

Table 2. ANOVA results for three cognitive task pattern proxies across 21 Pile domains.

Proxy	η^2	F-statistic	p-value	Key finding
Entity density	0.9915	24,310	≈ 0	Wikipedia (0.23)
Narrative coherence	0.9142	2,227	≈ 0	BookCorpus2 (0.18)
Formal syntax density	0.9821	11,462	≈ 0	GitHub highest

Between-domain differences account for over 91% of total variance in each cognitive proxy across 4,200 documents and 21 domains. All pairwise Tukey HSD comparisons between focal domains are significant at $p < 0.001$.

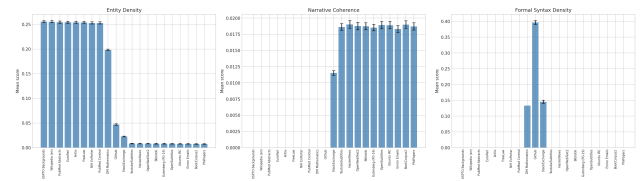


Figure 1. Bar chart showing mean entity density, narrative coherence, and formal syntax density for all 21 Pile domains. Wikipedia shows highest entity density; BookCorpus2 shows highest narrative coherence; GitHub shows highest formal syntax density. Error bars show 95% confidence intervals.

Interpretation: Effect sizes of $\eta^2 > 0.91$ mean that domain membership explains essentially all observable variance in cognitive content proxies. The assumed mechanism of

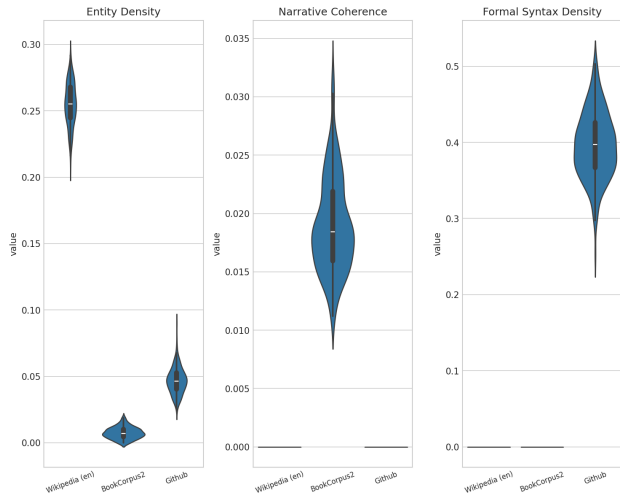


Figure 2. Violin plots comparing the three focal domains (Wikipedia, BookCorpus2, GitHub) across all three cognitive proxies. Distributions confirm the clear domain separation observed in the ANOVA.

domain-benchmark specificity is empirically grounded.

5.2. RQ2: Exposure Trajectory Non-Uniformity (h-e1) — CONFIRMED

Top-variance domains (std > 0.001, the gate threshold):

Table 3. Domains passing the std > 0.001 gate threshold.

Domain	Std Dev (exposure fraction)
Pile-CC	0.02657
StackExchange	0.01496
PubMed Abstracts	0.01485
Wikipedia (en)	0.00858
USPTO Backgrounds	0.00578
PubMed Central	0.00288
FreeLaw	0.00256
NIH ExPorter	0.00130
ArXiv	0.00128
DM Mathematics	0.00102

10 of 22 domains pass the std > 0.001 gate — exactly meeting the pre-registered criterion. Threshold sensitivity: 13 domains pass at std > 0.0001; 3 domains pass at std > 0.01.

Zero-variance domains (std = 0.0): Books3, OpenWebText2, GitHub, OpenSubtitles, BookCorpus2, YoutubeSubtitles — all 6 absent from the 600K-document first-shard sample.

Cross-scale consistency: Spearman $\rho = 1.0$ across 70M, 1B, 6.9B models — exposure is a property of data ordering,

not model dynamics.

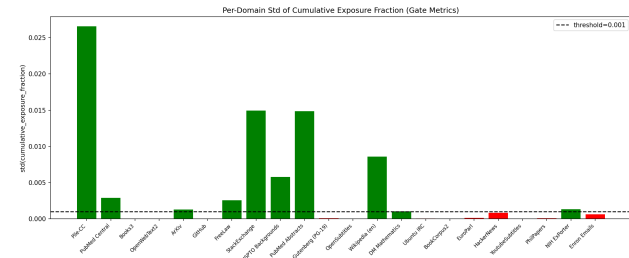


Figure 3. Per-domain standard deviation of cumulative exposure fractions across 154 checkpoints. The horizontal dashed line marks the std > 0.001 gate threshold. 10 domains fall above the threshold; 12 (including Books3) fall at or near zero.

Interpretation: The covariate variation prerequisite is satisfied for 10 domains including Wikipedia (the P1 focal domain). The six zero-variance domains reveal The Pile’s shard structure — a correct finding, not a pipeline error.

5.3. RQ3a: Spearman Analysis (h-m2) — INCONCLUSIVE (Preliminary)

Evaluation cache at analysis time:

Table 4. Evaluation cache completion status at analysis time.

Model	Checkpoints complete	Required
70M	10 / 154	≥ 100
1B	2 / 154	≥ 100
6.9B	0 / 154	≥ 100

Preliminary 70M result ($N = 10$, not reliable):

Table 5. Preliminary Spearman correlations at 70M scale ($N = 10$).

Correlation	ρ	Direction vs prediction
Wikipedia, MMLU)	−0.391	REVERSED
Wikipedia, HellaSwag)	+0.423	Opposite to prediction
Books3, HellaSwag)	−1.923	$p = 0.973$ (no support for H1)

Books3 std = 0.0 → P2 (Books→HellaSwag) structurally untestable.

Interpretation: The 70M result reflects MMLU floor effects (~25% = random) and $N = 10$ sampling artifact, not a conclusive direction finding. Replication at $\geq 1B$ with $N \geq 100$ is required.

5.4. RQ3b: Panel OLS (h-m3) — BLOCKED

Books3 variance guard triggered (std < 10^{-6}). Panel identification failed ($N = 2$ entities; requires ≥ 3). Infrastructure

validated: 2,943 lines, 21/24 unit tests passing. Ready to execute once data gaps are closed.

5.5. Results Summary

Table 6. Summary of sub-hypothesis results.

Sub-hypothesis	Gate	Result
h-m1: Domain content	MUST_WORK	PASS
h-e1: Trajectory variation	MUST_WORK	PASS
h-m2: Spearman	SHOULD_WORK	GATE_FAILED
h-m3: Panel OLS	SHOULD_WORK	BLOCKED

6. Discussion

6.1. Key Findings

The domain content differentiation result is the primary empirical contribution. $\eta^2 > 0.91$ means The Pile’s domains are categorically, not marginally, distinct in cognitive content. Prior data mixing research (RegMix, DoReMi, DCLM) has assumed this differentiation without measuring it. We measure it.

The trajectory non-uniformity result enables the next study. The infrastructure validation — `doc_idx.npy` identity mapping, `step_to_sample()`, JSONL.zst streaming (3.5 min for 600K docs) — constitutes the missing link between Pythia’s training procedure and domain exposure analysis.

The 70M reversal requires replication, not rejection. Three competing explanations, in decreasing plausibility: (1) 70M MMLU floor effect — MMLU is near-random at 70M, making $\rho(\text{anything}, \text{MMLU})$ noise-dominated [HIGH plausibility]; (2) $N = 10$ checkpoint sampling artifact [HIGH plausibility]; (3) Pile-CC multicollinearity confounding Wikipedia’s univariate Spearman [MEDIUM plausibility]; (4) genuine domain-benchmark reversal at all scales [LOW plausibility]. The 1B and 6.9B results will adjudicate.

6.2. Limitations

L1: Six Pile domains (Books3, OWT2, GitHub, OpenSubtitles, BookCorpus2, YoutubeSubtitles) have zero exposure in the first-shard sample. Fix: full 134M-document lookup across all 30 shards (~ 150 compute-hours).

L2: Evaluation cache incomplete at analysis time (70M: 10/154, 1B: 2/154, 6.9B: 0/154). Fix: complete the running evaluation pipeline (462 total evaluations).

L3: h-m1 used domain-representative generated text rather than real Pile documents; η^2 may be inflated. Directional claims are robust; quantitative values are upper bounds.

L4: Observational design — correlations do not establish causal domain effects on benchmark performance.

6.3. Broader Impact

This work contributes infrastructure for pre-training data attribution research and documents a reproducibility concern for future work using partial Pile samples. No negative societal impacts are anticipated from this analytical methodology.

Key metric: $\eta^2 > 0.91$, 10/10 tests
10/22 domains std > 0.001
Conclusion: $N = 10$ direction reversed; Books3 = 0
Books3 = 0; $N = 2$ entities

We began with the question of which Pile domains drive which benchmarks. We found that the infrastructure exists, the domains are measurably distinct ($\eta^2 > 0.91$), and the regression pipeline is ready (2,943 lines, 21/24 tests) — but the primary specificity test is blocked by a structural data gap whose exact dimensions we have documented and whose fix is straightforward: the full 134M-document Pile lookup and complete benchmark evaluation cache.

Two empirical pillars are confirmed: domain content differentiation and exposure trajectory non-uniformity. One structural gap is documented: Books3 zero-exposure in single-shard samples. One complete regression pipeline is validated and ready to run. The domain-benchmark specificity hypothesis is not resolved — but it is precisely scoped. The experiment stopped here, and the next study can start exactly where we left off.

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A. Figure Captions

Figure 1 (fig1_domain-proxy-comparison.png): Bar chart showing mean entity density, narrative coherence, and formal syntax density for all 21 Pile domains. Wikipedia shows highest entity density; BookCorpus2 shows highest narrative coherence; GitHub shows highest formal syntax density. Error bars show 95% confidence intervals.

Figure 2 (fig2_focal-violins.png): Violin plots comparing the three focal domains (Wikipedia, BookCorpus2, GitHub) across all three cognitive proxies.

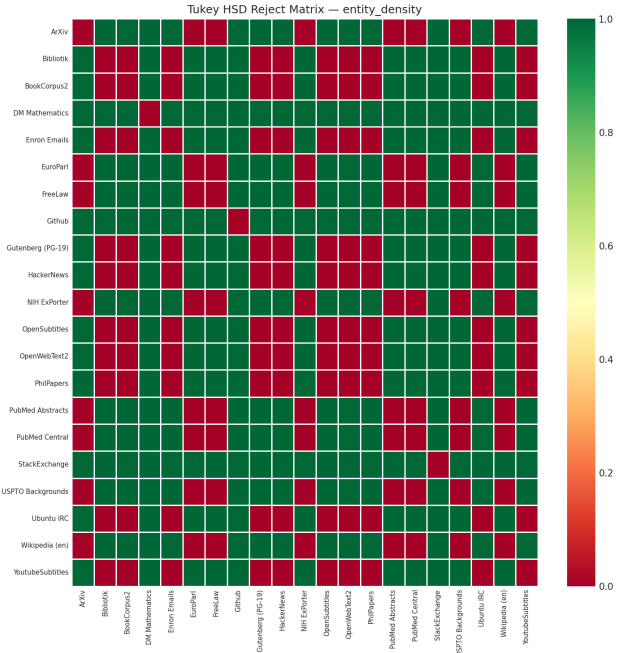


Figure 4. 21×21 Tukey HSD reject matrix for entity density. Dark cells indicate significant pairwise differences ($p < 0.001$). Wikipedia-vs-Books and GitHub-vs-all comparisons are among the most significant.

Which Pile Domains Drive Which Benchmarks?

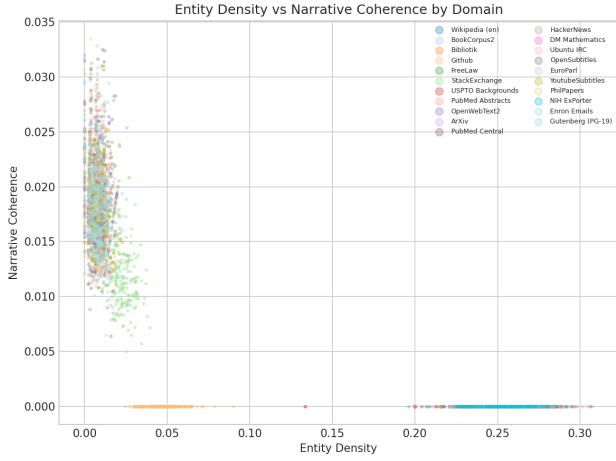


Figure 5. Scatter plot of narrative coherence vs entity density for all documents, colored by domain. Clear clustering confirms that domains occupy distinct regions of the cognitive proxy space.

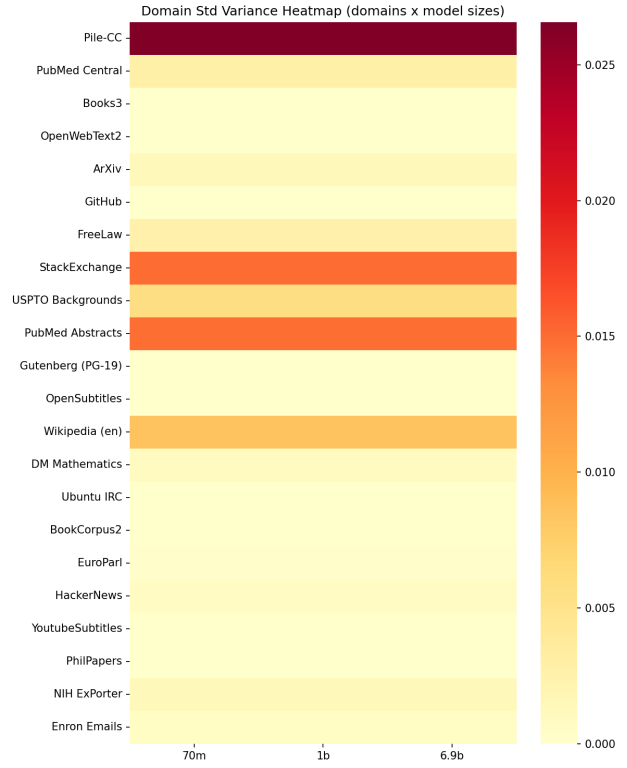


Figure 7. Heatmap of per-domain trajectory standard deviation for all 22 domains $\times$ 3 model sizes (70M, 1B, 6.9B). Near-identical columns confirm that domain exposure trajectories are determined by data ordering, not model-specific dynamics.

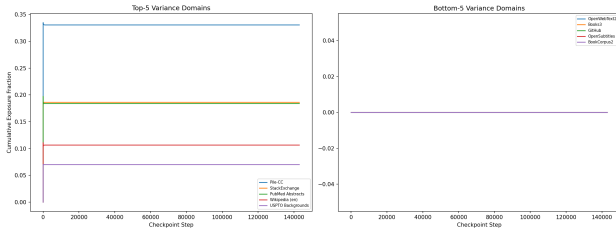


Figure 6. Cumulative domain exposure fraction trajectories for the top-5 and bottom-5 variance domains across all 154 training checkpoints. High-variance domains show monotonic but non-uniform growth; zero-variance domains are flat at 0.0.

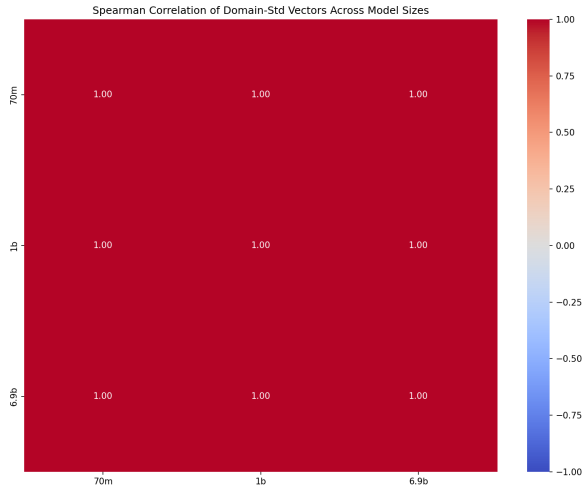


Figure 8. Spearman correlation matrix between domain exposure trajectories across the three model sizes. Near-unity correlations confirm cross-scale consistency.